

“CROP YIELD PREDICTION USING MACHINE LEARNING”

**by
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Guided Supervision

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ABSTRACT

- India is an agricultural nation and depends a lot on its agricultural activities.
- The purpose of this study is to apply the crop yield prediction approach into action to assist farmers in solving agricultural production concerns.
- We aim to predict the yield of various crops such as Rice, maize etc. Numerous studies have been conducted in the agricultural sector to better estimate crop yield through machine learning techniques.
- Machine Learning (ML) is the best technique which gives a better practical solution to crop yield problem. So the Random Forest algorithm which we decided to use to train our model to give high accuracy and best prediction.
- It is an efficient method for anticipating crop yields and determining which crops to cultivate.



INTRODUCTION

- Agriculture is the engine that drives the Indian economy.
- Agricultural yield is essential in ensuring global food security since it influences the amount of food that can be produced for consumption.
- Applying ML to agriculture may result in sustainable farming methods.
- This study primarily concentrates on the Random Forest which is applied to predict the yield of crops.



MACHINE LEARNING

- Machine learning (ML) is a type of artificial intelligence (AI) that enables machines to learn from data and improve over time. ML is used to create systems that can perform tasks without being explicitly programmed.

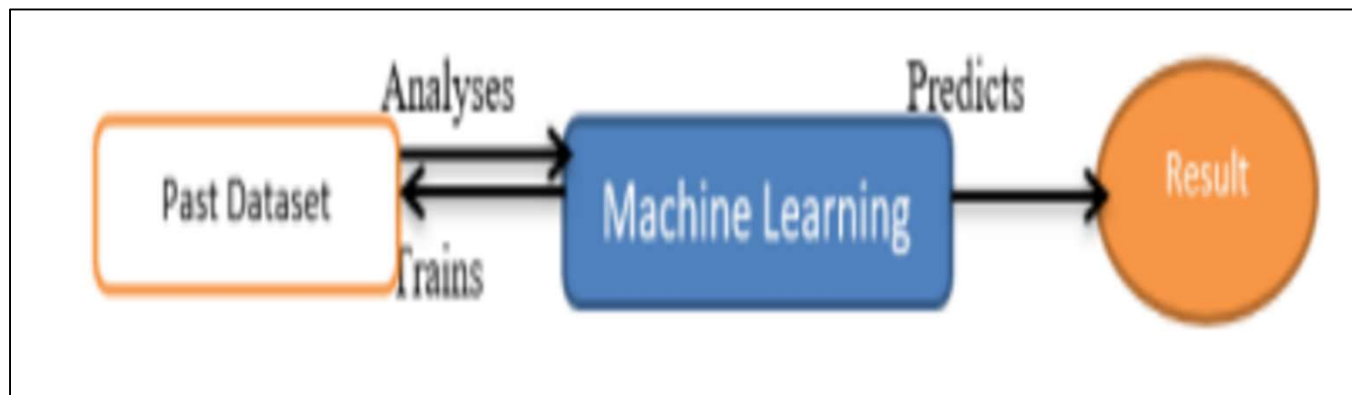
Types of ML models:

- **Supervised learning:** Models that make predictions after seeing data with the correct answers.
- **Unsupervised learning:** Models that make predictions by analyzing unlabeled data
- **Semi - Supervised learning:** SSL is a machine learning technique that uses both labeled and unlabeled data to train models.
- **Reinforcement learning:** Models that learn by getting rewards or penalties based on actions

Random Forest Algorithm

- Random Forest is a supervised machine learning algorithm that uses an ensemble of decision trees to make predictions, suitable for both classification and regression tasks, improving accuracy and stability compared to a single decision tree.

Process of Machine Learning:





OBJECTIVES

- The main objective of crop yield prediction is to improve decision-making in agriculture, optimize resource use, mitigate risks, and ultimately enhance food security and sustainability by accurately forecasting potential yields.
- To provide a solution for farmers decide which crop to grow in addition to increasing yield.
- Farmers can benefit from yield estimation because it allows them to minimize crop loss and obtain the best prices for their crops.



EXISTING SYSTEM

- In our research, which we found in the previous research papers is that everyone uses climatic factors like rainfall, sunlight and agricultural factors like soil type, nutrients possessed by the soil (Nitrogen, Potassium, etc.)
- But the problem is we need to gather the data and then a third party does this prediction and then it is explained to the farmer and this takes a lot of effort for the farmer and he doesn't understand the science behind these factors.



DISADVANTAGES

- Limited Data Availability
- Low Precision in Small Regions
- Limited Use of Advanced ML Techniques
- Poor Farmer Accessibility



PROPOSED SYSTEM

- The Proposed model is to predict the crop yield. We have collected the dataset for the crop prediction from real world sensor data available in **kaggle.com**.
- This dataset contains attributes such as soil, temperature, pH, Rainfall, Humidity.
- For getting high accuracy we used the Random Forest algorithm which gives accuracy which predicate by model and actual outcome of predication in the dataset.



ADVANTAGES

- Improved Decision-Making
- Increased Productivity
- Climate Adaptation
- Market and Price Forecasting
- Sustainability



SYSTEM REQUIREMENTS

- **HARDWARE REQUIREMENTS**

- ❖ Processor : Any Processor Above 1 GHz
- ❖ RAM : Minimum 4 GB
- ❖ Hard Disk : Few (5 GB) free Disk Space
- ❖ Input Device: Standard Keyboard and Mouse
- ❖ Output Device : A monitor with 1024x768 resolution is sufficient

- **SOFTWARE REQUIREMENTS**

- ❖ Operating System : Windows 7 or Higher
- ❖ Programming : Python is used within Google colab



MODULE

- The “CROP YIELD PREDICTION USING MACHINE LEARNING” module is a core component of an agriculture decision support system.
- It uses Machine Learning (ML) and data analytics to predict crop yields based on various factors, including weather patterns, soil properties, and farming practices.
- The Module List contains,
 - ❖ Data Preprocessing
 - ❖ Data Loading
 - ❖ Data exploration
 - ❖ Data Analysis through Visualization
 - ❖ Prediction



FEASIBILITY

- Crop yield prediction is highly feasible and its accuracy is continuously improving.
- This technology plays a vital role in modern agricultural, contributing to food security, resource management and risk mitigation



RESULTS

- We observed that the random forest technique performed better.
- According to results from experiments on crop yield data, the best outcome, with an accuracy value of 0.99, is produced by an improved Random Forest.
- Despite the high yield accuracy, this work still has certain shortcoming.
- The use of machine learning, particularly the Random Forest algorithm, is becoming increasingly prevalent in crop yield prediction.



FUTURE ENHANCEMENT

- Future enhancements for crop yield prediction using machine learning include leveraging advanced algorithms like deep learning (CNN, LSTM, DNN), incorporating more data sources (remote sensing, IoT), and focusing on real-time prediction and personalized recommendations for farmers.
- Implement real-time prediction systems that can provide timely insights and alerts to farmers.
- Incorporate XAI techniques to provide insights into the model's predictions, helping to build trust and understanding among users.



CONCLUSION

- The Conclusion of crop yield prediction is that it is mainly useful for the farmers who are facing some difficulties while cultivating their crops and who don't have idea about what crop to grow on certain conditions.
- So this model helps them to make a better decisions and it will increase the productivity and decrease the farmer's economy.



THANK YOU